

Deep MNIST Champion Project Report

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1. Project Overview

This project demonstrates the development of a high-performance Handwritten Digit Classifier using the MNIST dataset. The objective was to engineer a "Champion" model achieving **>99%** accuracy through professional MLOps practices, including rigorous hyperparameter tuning and state persistence.

2. Dataset Description

The MNIST dataset consists of 70,000 grayscale images of handwritten digits (0–9), each formatted as **28 × 28** pixels.

- **Training Set:** 60,000 images
- **Test Set:** 10,000 images

3. Model Architecture

The architecture utilizes a 3-Stage Convolutional Neural Network (CNN) optimized for deep feature extraction:

1. **Stage 1:** Conv2D + Batch Normalization + MaxPooling + Dropout (0.4)
2. **Stage 2:** Conv2D + Batch Normalization + MaxPooling + Dropout (0.3)
3. **Stage 3:** Conv2D + Batch Normalization
4. **Output:** Flatten + Dense (ReLU) + Softmax (10 classes)

4. Performance Results

Metric	Final Value
Training Accuracy	99.44%
Validation Accuracy	99.27%
Training Loss	0.0158
Validation Loss	0.0255

The minimal gap between training and validation metrics confirms the model's excellent generalization capabilities.

5. Resilience & Persistence

To ensure progress was never lost due to external factors, the following were implemented:

- **Model Checkpointing:** Automatic saving of the best performing weights.
- **CSV Logging:** Real-time recording of all epoch metrics.
- **Early Stopping:** Monitoring validation loss to prevent overfitting.

6. Conclusion

By moving from unstructured scripts to a modular repository, this project serves as a cornerstone for a professional data science portfolio. The next phase will apply these methods to the *10 Monkey Species Classification* project using AutoML.